

Challenges in a module based FPGA design flow

There are mainly two challenges faced in the creation of a module based FPGA design flow. First, creation of modules should be fast and simple. The so-created modules should contain as much relative placement and routing information as possible to provide reasonable speedup.

The second challenge is the absence of a framework for commercial FPGAs, which allows the design flow to be built upon. Performing experiments on actual FPGAs ensures a credible speedup and fast build times. "RapidSmith follows a new hard macro based design flow that, like software, would create and use pre-compiled modules (or hard macros) from a library that could be rapidly assembled, placed and routed for a rapid implementation," describe the project administrators in their demonstration paper. The tool, thereby, overcomes the above-listed challenges and offers an open source framework that makes such a design flow possible on Xilinx FPGAs.

Targeting academicians in FPGA CAD research

The tool is targeted at academicians involved in the research of FPGA computer-aided design (CAD). A basic understanding of programming in Java is a prerequisite for using this tool.

Again, users need to have basic knowledge about Xilinx FPGAs and XDL, but the detailed documentation of the tool available online will help them be at par with the required knowledge level. The tool is intended for the purpose of research

only and should not be used with commercial designs.

What RapidSmith offers

Having discussed the motivation behind and targeted users, let us now have a quick look at the various tools bundled in the software.

Design-analysis tools. If a design is given in XDL format, RapidSmith provides an easy-to-use platform for creating design-analysis tools. A design analyser that tabulates detailed resource utilisation statistics and design properties could also be created.

Design-creation tools. RapidSmith could be used to read and write XDL files. Apart from this, the software allows users to create a circuitry in XDL file format. APIs that create primitive instances customise these instances and even create logical connections between primitive instance pins contained in the tool.

Physical design tools.

Creation of physical design tools is very easy with the help of RapidSmith framework. An interactive hard macro placement tool can be built using RapidSmith.

The use of Java

One of the main priorities behind the development of RapidSmith software is its ease of use. With user-friendly features like an easy-to-use API and automated garbage collection, Java was chosen as the programming language for RapidSmith. The time needed for debug is greatly reduced and the researcher can focus on more fruitful research. The design and device packages in Java alleviate the challenges of using XDL and establish an API to interface with the ISE tool flow through XDL interface.

But many engineers

feel that the use of Java as a prototyping platform for FPGA CAD tools is a wrong choice. This is due to its reputation of being a memory hog and slow. But developers of RapidSmith are optimistic in this regard and feel that both speed and memory can be controlled to the point where this is not an issue.

Improving productivity of FPGA designs

RapidSmith is an XDL based rapid prototyping platform for research ideas and algorithms relating to low-level FPGA CAD tools. Hopefully, researchers around the globe will make use of the tool for research to develop its capabilities further and to provide new opportunities for research in FPGA CAD tools. **EFY**

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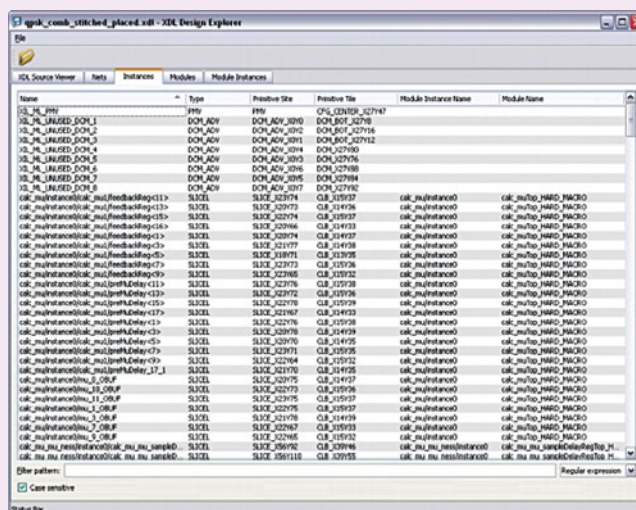


Fig. 3: A RapidSmith program that interactively searches XDL instances

Infobytes

Size of the tool: 345.5MB

Developed at: Brigham Young University, USA

Latest release: Version 0.5.2

Licence: GNU general-public licence

Supported operating systems (OSes): Windows, Linux, Mac OS X (RapidSmithworks; but Xilinx tools do not run on OS X)

Supported devices: Virtex 4 and 5 families; to a lesser extent other Xilinx families are also supported

Available at: <https://sourceforge.net/projects/rapidsmith/files/>